

Differentiation Techniques: The Power Rule and Sum-Difference Rules

Leibniz's Notation:

When y is a function of x , we will also designate the derivative, $f'(x)$, as

$$\frac{dy}{dx},$$

which is read “the derivative of y with respect to x .”

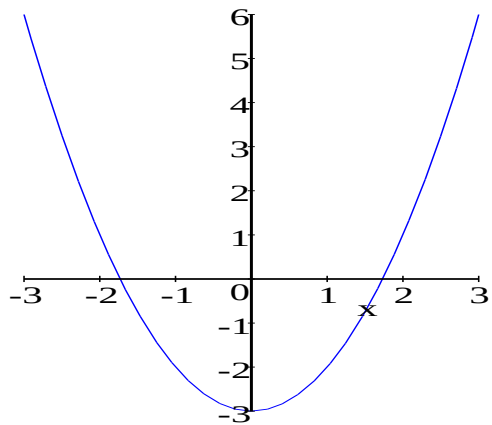
$f'(x)$ “f prime x” or “the derivative of f with respect to x”

y' “y prime”

$\frac{dy}{dx}$ “dee why dee ecks” or “the derivative of y with respect to x”

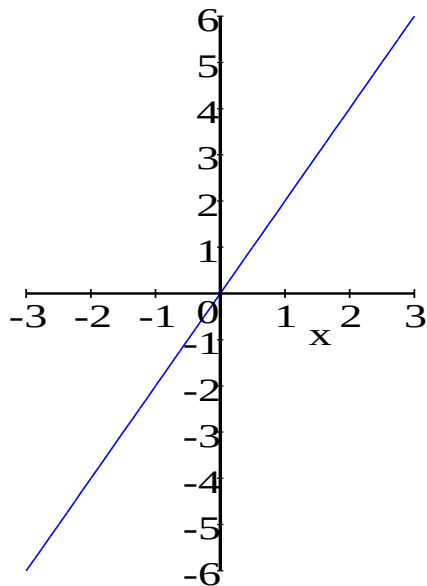
$\frac{df}{dx}$ “dee eff dee ecks” or “the derivative of f with respect to x”

$\frac{d}{dx} f(x)$ “dee dee ecks uv eff uv ecks” or “the derivative
 ($\frac{d}{dx}$ of f of x) of f of x”



$$y = x^2 - 3$$

$$y' = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 3 - (x^2 - 3)}{h}$$



$$y' = \lim_{h \rightarrow 0} 2x + h$$

$$y' = 2x$$

Derivatives of Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

Differentiation

[Chain Rule]

If $y \rightarrow u \rightarrow v \rightarrow w \rightarrow x$ then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dv} \times \frac{dv}{dw} \times \frac{dw}{dx}$$

Question # Find $\frac{dy}{dx}$ if $y = \sqrt{\sin \ln(ax^2 + b)}$

Solution:

We know $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dv} \times \frac{dv}{dw} \times \frac{dw}{dx}$

$$\begin{aligned} \therefore \frac{d}{dx} \sqrt{\sin \ln(ax^2 + b)} &= \frac{d\{\sin \ln(ax^2 + b)\}^{\frac{1}{2}}}{d\{\sin \ln(ax^2 + b)\}} \times \frac{d\{\sin \ln(ax^2 + b)\}}{d\{\ln(ax^2 + b)\}} \times \\ &\quad \frac{d\{\ln(ax^2 + b)\}}{d(ax^2 + b)} \times \frac{d(ax^2 + b)}{dx} \end{aligned}$$

$$= \frac{1}{2} \{\sin \ln(ax^2 + b)\}^{-\frac{1}{2}} \times \cos \ln(ax^2 + b) \times \frac{1}{ax^2 + b} \times 2ax$$

Differentiation

[Multiple and divide Rule]

$$\frac{d}{dx} \left(\frac{P(x)Q(x)R(x)}{S(x)T(x)} \right)$$
$$= \frac{P(x)Q(x)R(x)}{S(x)T(x)} \left\{ \frac{P'(x)}{P(x)} + \frac{Q'(x)}{Q(x)} + \frac{R'(x)}{R(x)} - \frac{S'(x)}{S(x)} - \frac{T'(x)}{T(x)} \right\}$$

Question # Find $\frac{dy}{dx}$ if $y = \frac{e^x \sin 2x \ln(3x)}{x^3 \cos(4x)}$

Solution:

We know

$$\frac{d}{dx} \left(\frac{P(x)Q(x)R(x)}{S(x)T(x)} \right) = \frac{P(x)Q(x)R(x)}{S(x)T(x)} \left\{ \frac{P'(x)}{P(x)} + \frac{Q'(x)}{Q(x)} + \frac{R'(x)}{R(x)} - \frac{S'(x)}{S(x)} - \frac{T'(x)}{T(x)} \right\}$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{d}{dx} \left\{ \frac{e^x \sin 2x \ln(3x)}{x^3 \cos(4x)} \right\} \\ \therefore &= \frac{e^x \sin 2x \ln(3x)}{x^3 \cos(4x)} \left\{ \frac{e^x}{e^x} + \frac{\sin(2x)}{2 \cos(2x)} + \frac{\ln(3x)}{\frac{1}{x}} - \frac{3x^2}{x^3} - \frac{-4 \sin(\pi)}{x^3 \cos(4x)} \right\} \end{aligned}$$

Question # Find $\frac{dy}{dx}$ if $y = x^x + \sin x^{\cos x}$

Let $u = x^x$ and $v = \sin x^{\cos x}$

$$\therefore y = u + v$$

$$\text{So, } \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

Now, $u = x^x$

$$\ln u = \ln x^x$$

$$\ln u = x \ln x$$

$$\therefore \frac{d}{dx} \ln u = \frac{d}{dx} x \ln x$$

$$\therefore \frac{d}{dx}(\ln u) = \frac{d}{dx}(x \ln x)$$

$$\text{Or, } \frac{1}{u} \times \frac{du}{dx} = \frac{1}{x} x + \ln x \times 1$$

$$\frac{du}{dx} = u(1 + \ln x)$$

$$\text{i.e. } \frac{du}{dx} = x^x(1 + \ln x)$$

Now again, $v = \sin x^{\cos x}$

$$\ln v = \ln \sin x^{\cos x}$$

$$\ln v = \cos x \ln \sin x$$

$$\therefore \frac{d}{dx} \ln v = \frac{d}{dx} (\cos x \ln \sin x)$$

$$\frac{1}{v} \times \frac{dv}{dx} = \cos x \times \frac{1}{\sin x} \times \cos x + \ln \sin x (-\sin x)$$

$$\frac{dv}{dx} = v(\cot x \cdot \cos x - \sin x \cdot \ln \sin x)$$

$$\frac{dv}{dx} = \sin x^{\cos x} (\cot x \cdot \cos x - \sin x \cdot \ln \sin x)$$

$$\text{So, } \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$= x^x (1 + \ln x) + \sin x^{\cos x} (\cot x \cdot \cos x - \sin x \cdot \ln \sin x)$$

Question # Find $\frac{dy}{dx}$ if $x^y + y^x = a$

Solution:

$$\text{Let } f(x, y) = x^y + y^x - a$$

$$\text{We know that, } \frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$$

Now, differentiating partially w. r. t. x , we get

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} x^y + \frac{\partial}{\partial x} y^x - \frac{\partial}{\partial x} a$$

$$\frac{\partial f}{\partial x} = yx^{y-1} + y^x \ln y$$

Similarly,
$$\frac{\partial f}{\partial y} = xy^{x-1} + x^y \ln x$$

$$\therefore \frac{dy}{dx} = - \frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = - \frac{yx^{y-1} + y^x \ln y}{xy^{x-1} + x^y \ln x}$$

Question

Compute the slope of the tangent to the curve $(y)^x + (x)^y + \ln(xy) = (x^3 + y^3)$ at the point $(1,1)$.

Solution:

Let $f(x, y) = (y)^x + (x)^y + \ln(xy) - (x^3 + y^3)$

We know that,
$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$$

Now, differentiating partially w. r. t. x , we get

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(y)^x + \frac{\partial}{\partial x}(x)^y + \frac{\partial}{\partial x}\ln(xy) - \frac{\partial}{\partial x}(x^3) - \frac{\partial}{\partial x}(y^3)$$

$$\frac{\partial f}{\partial x} = y^x \ln y + yx^{y-1} + y \ln(xy) - 3x^2 - 0$$

Similarly,
$$\frac{\partial f}{\partial y} = x^y \ln x + xy^{x-1} + x \ln(xy) - 0 - 3y^2$$

$\therefore$ the slope of the tangent to the curve ,

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{y^x \ln y + yx^{y-1} + y \ln(xy) - 3x^2}{x^y \ln x + xy^{x-1} + x \ln(xy) - 0 - 3y^2}$$

∴ the slope of the tangent to the curve at (1,1) ,

$$\begin{aligned}\left. \frac{dy}{dx} \right|_{(1,1)} &= - \frac{y^x \ln y + yx^{y-1} + y \ln(xy) - 3x^2}{x^y \ln x + xy^{x-1} + x \ln(xy) - 0 - 3y^2} \\ &= - \frac{0 + 1 + 0 - 3}{0 + 1 + 0 - 3} \\ &= -1\end{aligned}$$

Question # Find $\frac{dy}{dx}$ if $e^{xy} = (x^2 + y^2)$

Solution:

$$\text{Let } f(x, y) = e^{xy} - x^2 - y^2$$

$$\text{We know that } \frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$$

Now, differentiating partially w. r. t. x, we get

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(e^{xy}) - \frac{\partial}{\partial x}x^2 - \frac{\partial}{\partial x}y^2$$

$$\frac{\partial f}{\partial x} = ye^{xy} - 2x$$

$$\text{Similarly, } \frac{\partial f}{\partial y} = xe^{xy} - 2y$$

$$\therefore \frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{ye^{xy} - 2x}{xe^{xy} - 2y}$$

Q Differentiate $y = \cos^{-1} \frac{1-x^2}{1+x^2}$ with respect to $\tan^{-1} \frac{2x}{1-x^2}$

Solution:

$$\begin{aligned} y &= \cos^{-1} \frac{1-x^2}{1+x^2} \\ &= \cos^{-1} \frac{1-\tan^2 \theta}{1+\tan^2 \theta} \quad [\because x = \tan \theta \rightarrow \therefore \theta = \tan^{-1} x] \end{aligned}$$

$$= \cos^{-1} \cos 2\theta$$

$$= 2\theta$$

$$\therefore y = 2 \tan^{-1} x$$

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

Again, $z = \tan^{-1} \frac{2x}{1-x^2}$

$$= \tan^{-1} \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$= \tan^{-1} \tan 2\theta$$

$$= 2\theta$$

$$\therefore z = 2 \tan^{-1} x$$

$$\frac{dz}{dx} = \frac{2}{1+x^2}$$

$$\therefore \frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}} = \frac{\frac{2}{1+x^2}}{1+x^2} = 1$$

Q. # Differentiate $\tan^{-1} \frac{2x}{1-x^2}$ with respect to $\sin^{-1} \frac{2x}{1+x^2}$